Name of the module: **LOG**
Location in GIT repository: vspa_common\vspa-lib\log

Overview:

This module provides a kernel to compute a general purpose logarithm function:

$$y = f(x) = fact * log2(x)$$

where *log2()* is the built-in Matlab function. The following cases can be used:

- $fact = 1$ for $f(x) = log2(x)$
- $fact = 10 * log10(2)$ for $f(x) = 10 * log10(x)$
- $fact = 20 * log10(2)$ for $f(x) = 20 * log10(x)$

Type of code:

- VCPU assembly function:
  - `log_asm`

Assembly Functions API:

```
typedef enum {
    LOG2x1   = 0x3f800000,      // Logarithm factor to compute f(x) = log2(x)
    LOG10x10 = 0x4040A8C1,      // Logarithm factor to compute f(x) = 10 * log10(x)
    LOG10x20 = 0x40c0a8c1,      // Logarithm factor to compute f(x) = 20 * log10(x)
} LOG_FACT_T;

float32_t log_asm(float32_t x, LOG_FACT_T fact);
```

x – Input value (32-bit floating point Single Precision)
fact – Logarithm factor for desired function (value chosen from the enumeration type `LOG_FACT_T`)
The function returns the logarithm function in 32-bit floating point Single Precision.

Implementation plan

The implementation uses the VSPA hardware instruction *log2* to compute other log functions. The output format of the log2 function is 32-bit fix-point with 1 sign bit, 16 bits integer part and 15 bits fractional part. The kernel mainly computes the following expression (with extra conversions from fix-point to floating point):

$$out = fact * [log2()] + fact * \{log2()\}$$

where [ ] and { } denote the integer part and the fractional part respectively.

The approximation error for each case is presented below. The approximation error is taken as the maximum absolute error between the Matlab double precision computation and the VSPA model:

Logarithm function	Approximation error
$f(x) = log2(x)$	1.5923e-04
$f(x) = 10 * log10(x)$	4.8876e-04
$f(x) = 20 * log10(x)$	9.7753e-04

DMEM requirements

No DMEM is used other than the API parameters. No scratch/rom/persistent memory is used.

Test plan:

- Tested features: test logarithm function for all factors.
- Take a look in **Log_Testplan.xlsx** for test plan details
- Take a look in **Log_Report.csv** for cycle count